



French University in Armenia

Faculty of Computer Science and Applied Mathematics

Internship Report

Master 1 - Artificial Intelligence

Synthetic PET Image Generation Using GAN and Conditional GAN for Medical Data Augmentation

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I am grateful to IRIT and to the academic framework involving UFAR and Université de Toulouse for providing the research context in which this internship was carried out. This internship allowed me to deepen my understanding of medical image synthesis, PET/CT data processing, and generative deep learning models.

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Résumé

La Tomographie par Émission de Positons (PET) est une modalité d'imagerie fonctionnelle utilisée en oncologie, mais les jeux de données PET annotés restent limités en raison du coût d'acquisition, de la sensibilité des données patient et de l'expertise nécessaire pour l'annotation tumorale. Ce stage s'inscrit dans ce contexte et étudie la génération d'images PET synthétiques comme stratégie d'augmentation de données.

Le travail porte sur l'analyse et l'implémentation de modèles génératifs, en particulier des Generative Adversarial Networks (GANs) et des Conditional Generative Adversarial Networks (cGANs). L'objectif est d'étudier comment différentes informations médicales, telles que les masques tumoraux et les images CT, peuvent guider la génération d'images PET synthétiques plus cohérentes structurellement.

Ce rapport présente le contexte scientifique du stage, l'état de l'art, la méthodologie mise en place, les difficultés rencontrées, ainsi que l'analyse des résultats obtenus. Il met également en évidence les limites actuelles de la génération PET synthétique et les perspectives nécessaires avant une éventuelle utilisation comme outil d'augmentation de données.

Abstract

Positron Emission Tomography (PET) is a functional imaging modality used in oncology, but annotated PET datasets remain limited due to acquisition cost, patient-data sensitivity, and the need for expert tumor annotation. This internship addresses this context by studying synthetic PET image generation as a data augmentation strategy.

The work focuses on the analysis and implementation of generative models, especially Generative Adversarial Networks (GANs) and Conditional Generative Adversarial Networks (cGANs). The objective is to study how different medical inputs, such as tumor masks and CT images, can guide the generation of more structurally coherent synthetic PET images.

This report presents the scientific context of the internship, the state of the art, the implemented methodology, the difficulties encountered, and the analysis of the obtained results. It also highlights the current limitations of synthetic PET generation and the perspectives required before its possible use as a data augmentation tool.

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1. Introduction

Deep learning has become a major research direction in medical image analysis, particularly for diagnostic support, reconstruction, image synthesis, and segmentation. However, its performance depends on large, diverse, and well-annotated datasets, which are difficult to obtain in medical imaging because of clinical availability, privacy constraints, acquisition cost, and expert annotation.

This limitation is especially relevant in PET imaging. PET provides functional information about metabolic activity and is valuable for oncology and tumor-related analysis, but PET datasets are often small and difficult to annotate at scale. While basic data augmentation techniques such as rotation, flipping, cropping, scaling, and intensity transformations can increase the apparent dataset size, they mostly modify existing images. In medical imaging, this is often insufficient because clinically meaningful anatomical and pathological variability must be preserved.

For this reason, synthetic medical image generation has become an important research direction. Instead of applying only simple transformations, generative models aim to create new image samples that follow the distribution of real medical data. In this internship, this idea was studied in the context of synthetic PET generation, with particular attention to whether generated images can preserve anatomical and lesion-related consistency.

The report first presents the internship environment and the assigned mission, then introduces the state of the art on medical image synthesis and PET generation. It then describes the implemented methodology, analyzes the obtained results, discusses the difficulties encountered, and concludes with perspectives for future work.

2. Host Institution and Internship Environment

2.1 Host institution

The internship was carried out within the framework of the Institut de Recherche en Informatique de Toulouse (IRIT), a research laboratory dedicated to computer science and digital technologies. IRIT's scientific activity covers a wide range of research areas in information technology, including artificial intelligence, data analysis, image and signal processing, interaction, decision support, modelling, systems, networks, and software reliability. Its research is organized around major scientific questions such as the design of reliable and adaptive systems, the modelling of real-world digital phenomena, the transformation of raw data into intelligible information, and the development of autonomous systems able to adapt to their environment.

Within this environment, my internship was connected to the MINDS team, whose name refers to CoMputational imagINg anD viSion. The team's research orientation is closely related to computational imaging, computer vision, image analysis, and data-driven visual interpretation. This scientific context was directly aligned with my internship topic, since the project required the analysis and generation of medical images using deep learning models. The work therefore fit naturally within research directions involving image modelling, visual information extraction, and artificial intelligence applied to complex data.

The internship was supervised by Amel Aissaoui on the host-laboratory side and by Kristina Sargsyan on the academic side. The combination of host supervision and academic supervision helped maintain both scientific relevance and alignment with university expectations.

2.2 Scientific and professional context

The scientific context of the internship was the need for robust AI methods in medical imaging despite limited available data. In natural image processing, large public datasets often make deep learning experimentation easier. In medical imaging, this is rarely the case. Acquisition cost, confidentiality, annotation difficulty, and the rarity of some pathologies restrict the amount of available data.

The project therefore addressed a relevant research problem: how generative models can be used to create additional PET-like image data while preserving medical consistency. The work was not

intended to create a clinical tool ready for use. Instead, it aimed to investigate the feasibility and limitations of synthetic PET generation approaches.

The work involved reading articles, implementing notebooks, debugging training behavior, preparing visual results, and explaining technical choices in a rigorous way.

2.3 Tools and working environment

The implementation was developed mainly in Python using PyTorch. Google Colab was used for GPU execution and iterative experimentation. Medical images were processed from volumetric NIfTI files using nibabel. The analysis and visualization were performed with matplotlib, NumPy, and standard scientific Python tools.

Element	Description
Programming language	Python
Deep learning framework	PyTorch
Execution environment	Visual Studio Code with GPU support
Medical image format	NIfTI (.nii.gz)
Image processing and numerical tools	NumPy, matplotlib, torchvision transforms
Main deliverables	Notebooks, generated images, loss curves, metrics, report, poster, GitHub repository

2.4 Confidentiality and data handling

The project used medical imaging data and therefore required careful treatment of confidentiality. The dataset itself is not included in public deliverables or the GitHub repository. Only code, selected figures, and references are intended to be shared. This decision respects the confidentiality and licensing constraints typically associated with medical datasets.

3. Internship Mission and Objectives

3.1 Problem addressed during the internship

The main problem addressed during the internship was the limited availability of PET image data for deep learning-based medical image analysis. PET images are valuable because they provide functional information, but their acquisition and annotation are expensive and constrained. Traditional data augmentation can increase the number of samples by applying transformations, but it does not create truly new anatomical or pathological patterns.

The internship therefore investigated synthetic PET image generation as a more advanced data augmentation strategy. The technical question was whether GAN-based and cGAN-based models could generate PET-like images that are not only visually plausible, but also structurally coherent with medical information such as tumor masks and CT images.

3.2 Objectives

- Review the scientific literature on medical image synthesis, PET image generation, GANs, and conditional GANs.
- Implement a standard GAN baseline for PET image generation.
- Implement conditional GAN models with different conditioning signals: mask only, CT only, and CT + mask.
- Compare the visual and quantitative behavior of different conditioning strategies.
- Analyze the role of preprocessing parameters, especially image size and minimum mask-pixel threshold.
- Prepare research deliverables: notebooks, figures, metrics, report, poster, and selected references.

3.3 Personal contribution

My personal contribution consisted of reading and synthesizing the relevant literature, developing the notebooks, designing the GAN and cGAN workflows, implementing several conditioning strategies, producing visual outputs and loss curves, evaluating the models, and preparing the report and poster. I also analyzed the limitations of each model and used these failures to guide the next implementation step.

3.4 Expected outcome

The expected outcome was not a clinically validated model, but a research prototype and methodological comparison. The aim was to identify which generative strategy is most promising for future synthetic PET generation and to understand which conditions provide useful information for PET-like image synthesis.

4. State of the Art (Bibliography)

4.1 Data scarcity and traditional augmentation

Deep learning models generally benefit from large and diverse datasets. In medical imaging, however, data availability is limited by acquisition cost, privacy constraints, and the need for expert annotation. Traditional data augmentation methods, such as rotation, flipping, cropping, scaling, and intensity transformations, are commonly used to reduce the effect of small datasets.

These classical augmentation techniques are useful, but they mostly modify existing images. For tumor-related medical imaging, this limitation is important because simple transformations do not necessarily create new clinically meaningful anatomical or pathological patterns. This is one reason why synthetic medical image generation has become an active research direction.

4.2 Deep learning-based medical image synthesis

The literature review by Sherwani and Gopalakrishnan[2] presents deep learning-based medical image synthesis as a broad research area, with applications in MRI-to-CT synthesis, CBCT-to-CT synthesis, low-dose to high-dose image generation, and PET attenuation correction. This shows that synthetic generation is not only a data augmentation strategy; it can also support modality translation, denoising, reconstruction, and clinical workflows when one imaging modality is missing or difficult to acquire.

The reviewed literature identifies CNNs, U-Net-like models, GANs, cGANs, and CycleGANs as important architectures. These methods differ in their training strategy and input-output relationship. For this internship, this distinction was important because the task progressively moved from unconditional generation toward image-to-image conditional synthesis.

4.3 GAN-based PET image generation

For the task closest to this internship topic, namely synthetic PET generation and PET reconstruction, the reviewed literature mainly supports GAN-based approaches. Islam and Zhang[3] demonstrated GAN-based synthetic brain PET image generation, showing that GANs can be applied directly to PET data. Xue et al. also showed that adversarial learning can be used for PET reconstruction through LCPR-Net.

These works motivated the initial GAN baseline in this internship. A standard GAN is a natural starting point because it learns the distribution of real PET images and generates new samples from random noise. However, this also creates a limitation: the model does not explicitly know where the lesion is located or what anatomical context surrounds it.

For medical data augmentation, this limitation is important. A visually plausible PET image is not sufficient if the generated image is not coherent with tumor-related structure. Therefore, while GAN-based PET literature supports the feasibility of synthetic PET generation, it does not fully solve the problem of tumor-structure-aware PET augmentation.

4.4 Conditional generation in medical imaging

Conditional GANs extend the GAN framework by adding conditioning information to the generator and discriminator. The condition can be a label, an input image, a segmentation mask, or another imaging modality. In the reviewed literature, cGANs appear mainly in related medical image synthesis tasks rather than directly in tumor-guided PET augmentation.

For example, conditional adversarial models have been used for denoising, attenuation correction, low-dose to full-dose PET estimation, CT-to-PET synthesis, and image-to-image translation. Ben-Cohen et al.[6] showed that CT information can guide PET-like synthesis for lesion detection, while pix2pix-style image-to-image translation provides a general framework for paired conditional generation.

These works motivated the transition from standard GAN generation toward conditional PET generation in this internship. In the implemented models, the conditions were a tumor mask, a CT slice, or the combination of CT and mask. The goal was to test whether medical conditions could improve structural coherence compared with uncontrolled generation.

4.5 Gap identified from the literature

The main gap identified was that PET synthesis is promising but remains less developed than MRI or CT synthesis. In addition, PET-related GAN studies often focus on general generation, reconstruction, or modality translation, rather than tumor-structure-aware generation for data augmentation.

The literature therefore motivated a progressive methodology: first, a GAN baseline was implemented because GANs are the most direct reference for PET generation; then, cGAN models

were tested because related medical image synthesis studies show that conditioning can guide generation using medical information. This supported the direction of comparing mask-only, CT-only, and CT + mask conditioning for synthetic PET generation.

5. Methodology (Appendix H)

5.1 Preliminary GAN/cGAN implementation on Fashion-MNIST

Before moving to PET data, standard GAN and cGAN models were first implemented on Fashion-MNIST. This preliminary stage was used to understand the basic adversarial training process in a simpler image-generation setting and to compare unconditional generation with conditional generation.

In the standard GAN, the generator receives a random latent vector and produces a synthetic image, while the discriminator learns to distinguish real images from generated ones. Architecturally, the generator projects the latent vector into a low-resolution feature representation and progressively upsamples it into a grayscale image. The discriminator follows the opposite direction: it receives an image, extracts features through convolutional layers, and outputs a real/fake decision.

In the cGAN version, class labels are added as conditioning information. The Fashion-MNIST labels correspond to ten clothing categories, such as T-shirt/top, Trouser, Sandal, Sneaker, Bag, and Ankle boot. In the generator, the class label is converted into an embedding and concatenated with the random noise vector, so the model learns to generate an image corresponding to the requested class. In the discriminator, the label is also embedded and combined with the image, allowing the discriminator to evaluate not only whether the image looks real, but also whether it matches the given class condition.

This preliminary implementation clarified the difference between uncontrolled generation and condition-guided generation. In the medical setting, the same idea of conditioning was later adapted, but the condition changed from a simple class label to medical information such as tumor masks or CT images.

5.2 PET/CT data preparation

The PET/CT dataset consisted of 33 original 3D patient volumes, corresponding to 11,185 slices. Since the implemented models operate on 2D images, the 3D volumes were converted into axial 2D slices. Each training sample was composed of aligned PET, CT, and tumor-mask slices.

A first difficulty was that many slices contained no tumor region or only a very small tumor area. Such slices are not useful for learning tumor-guided PET generation because the condition contains little or no lesion information. To address this, a minimum tumor-mask-pixel threshold of 200 was applied. Slices containing fewer than 200 tumor-mask pixels were excluded, leaving 676 valid slice pairs. The dataset was then divided into 70% training, 20% validation, and 10% testing.

Another important issue was resizing. PET and CT images contain continuous intensity values, while tumor masks are binary. In early preprocessing attempts, using ordinary interpolation for masks could blur lesion boundaries and introduce intermediate values between 0 and 1. This would make the mask less reliable as a condition. To solve this, nearest-neighbor interpolation was used for masks, while PET and CT were resized as continuous intensity images. This preserved the binary nature of the lesion masks and improved the consistency of the conditioning input.

5.3 Standard PET GAN baseline

The first PET model was a standard GAN baseline. In this configuration, the generator receives a random latent vector and produces a synthetic PET slice. The discriminator receives either a real PET slice or a generated PET slice and predicts whether the image is real or fake.

The generator follows a progressive upsampling design: a latent vector is transformed into a compact feature representation and then expanded into a PET-sized image. The discriminator applies convolutional layers to extract features from PET slices and produces a real/fake decision. This model was selected as the simplest GAN-based reference because several reviewed PET-generation works use GAN-based synthesis.

The main limitation encountered with this approach was methodological: the model had no access to medical structure. Since it receives only random noise, it does not know tumor location, lesion shape, or anatomical context. This made the standard GAN useful as a baseline, but not sufficient as a final approach for tumor-related PET generation. This limitation motivated the transition toward conditional models, where the generation process could be guided by medical information.

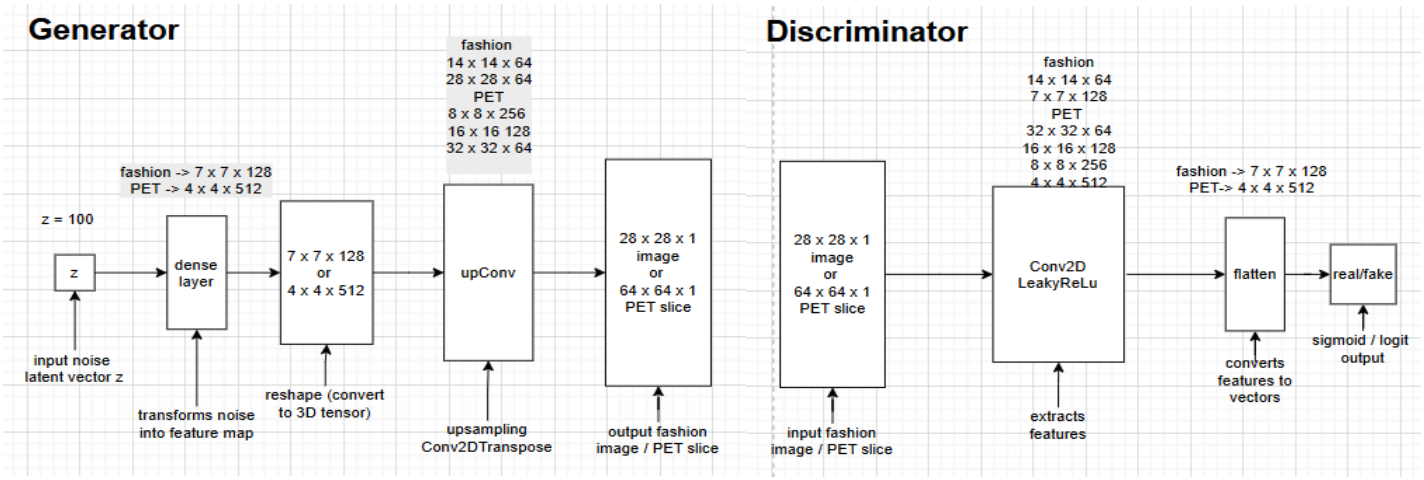


Figure 1. Standard GAN architecture used as the baseline for PET image generation.

5.4 PET cGAN with mask conditioning

The first conditional PET model used the tumor mask as input condition. The generator receives a binary tumor mask and produces a synthetic PET slice. The discriminator receives the mask concatenated with either a real or generated PET slice. This changes the task from random generation to paired image-to-image generation.

The generator was designed as an encoder-decoder network. The encoder reduces spatial resolution while increasing feature depth, allowing the model to extract information about lesion position and shape from the mask. The decoder reconstructs a PET-like image from this representation. This architecture was chosen because the mask and the target PET slice are spatially aligned: the lesion location in the mask should correspond to the PET activity in the output.

A difficulty in this model was that adversarial loss alone was not enough. A generated image could appear visually plausible but still be poorly aligned with the paired real PET target. To address this, the generator objective combined adversarial Binary Cross-Entropy loss with L1 reconstruction loss:

$$\mathcal{L}_{\mathrm{G}} = \mathcal{L}_{BCE}(D(x, G(x)), 1) + \lambda \|G(x) - y\|_1$$

where $x$is the conditioning input (mask, CT, or CT + mask), $y$is the real PET target, and $G(x)$is the generated PET image. The BCE term encourages realistic generation, while the L1 term

preserves spatial similarity with the real PET image. The parameter λ controls the weight of the reconstruction loss.

A second limitation was that the mask condition only provides lesion location and shape. It does not contain anatomical information about the surrounding body structure. This motivated the later use of CT conditioning.

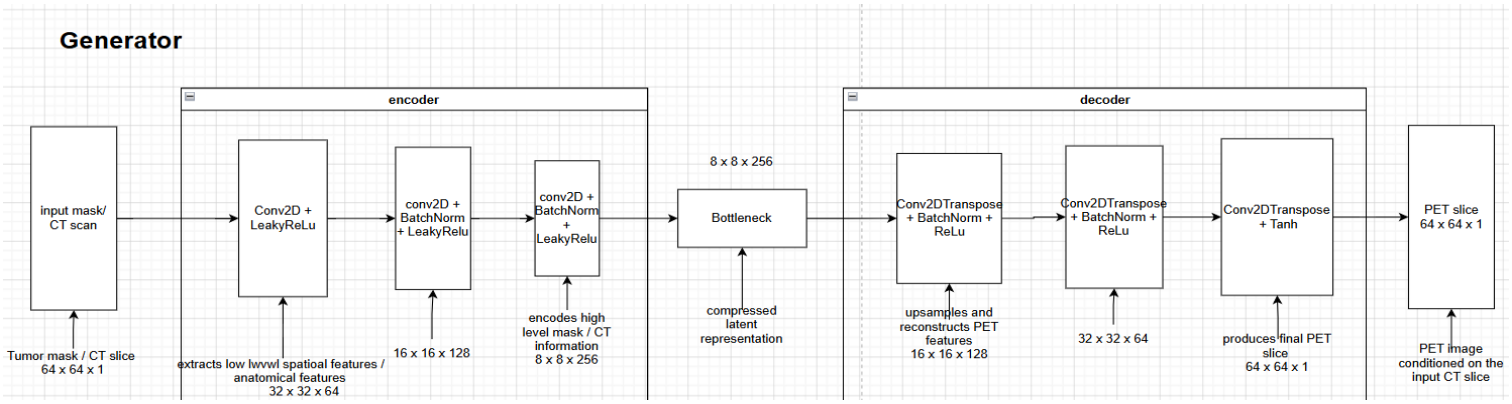


Figure 2. Generator architecture for mask-conditioned PET cGAN.

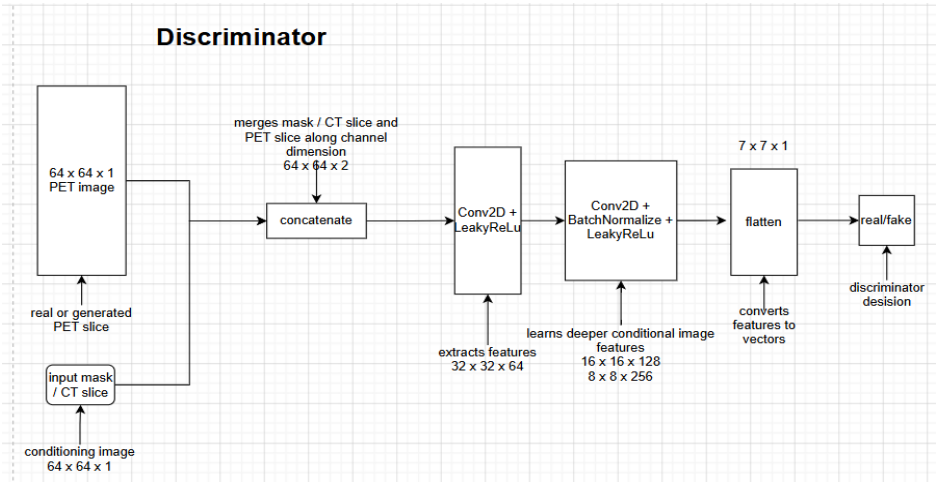


Figure 3. Discriminator architecture for mask-conditioned PET cGAN.

5.5 Checkerboard artifacts and decoder correction

A major implementation issue was the appearance of grid-like/checkerboard artifacts in generated PET images. These artifacts appeared as systematic texture patterns and made the output less reliable visually. In medical image generation, this is especially problematic because artificial patterns may be confused with PET uptake or anatomical texture.

This issue was linked to the upsampling strategy used in the generator decoder. Early versions relied on transposed convolution or unsuitable resizing choices. Transposed convolution can create uneven overlap during upsampling, which may lead to checkerboard artifacts. To reduce this problem, the decoder was revised to use explicit upsampling followed by convolution:

upsampling -> convolution -> batch normalization -> activation

This separates the resizing step from feature learning. It helped reduce systematic grid artifacts and produced smoother PET outputs. In parallel, nearest-neighbor interpolation was kept for binary masks to avoid blurred tumor boundaries during preprocessing.

5.6 PET cGAN with CT conditioning

The second conditional PET model used the CT slice as input condition. The generator receives a CT slice and generates a corresponding synthetic PET slice. The discriminator receives the CT slice concatenated with either a real or generated PET slice.

The architecture follows the same encoder-decoder principle as the mask-conditioned cGAN, but the condition has a different meaning. The mask provides explicit lesion localization, while CT provides anatomical context. In this model, the encoder extracts anatomical features from the CT slice, and the decoder transforms these features into a PET-like output.

The main implementation question was whether anatomical information alone could guide PET generation. CT contains structural information, but it does not explicitly indicate the tumor region or metabolic uptake. Therefore, this model was implemented to isolate the effect of anatomical conditioning and compare it with lesion-based conditioning.

5.7 PET cGAN with CT and mask conditioning

The final conditional PET model combined CT and tumor-mask information. In this configuration, the generator receives a two-channel condition composed of the CT slice and the tumor mask. The output is a synthetic PET slice. The discriminator receives the CT slice, the mask, and either a real or generated PET slice concatenated together.

This architecture was designed to combine complementary medical information. The CT channel provides anatomical context, while the mask channel provides explicit lesion-location information. The generator processes both channels jointly in the encoder and reconstructs the PET image through the decoder. The discriminator evaluates whether the generated PET slice is realistic under both conditions.

This model was introduced to address the limitations of the single-condition approaches. Mask-only conditioning gives lesion location but lacks anatomy; CT-only conditioning gives anatomy but lacks explicit lesion localization. Combining the two conditions was therefore the most complete strategy from a methodological perspective.

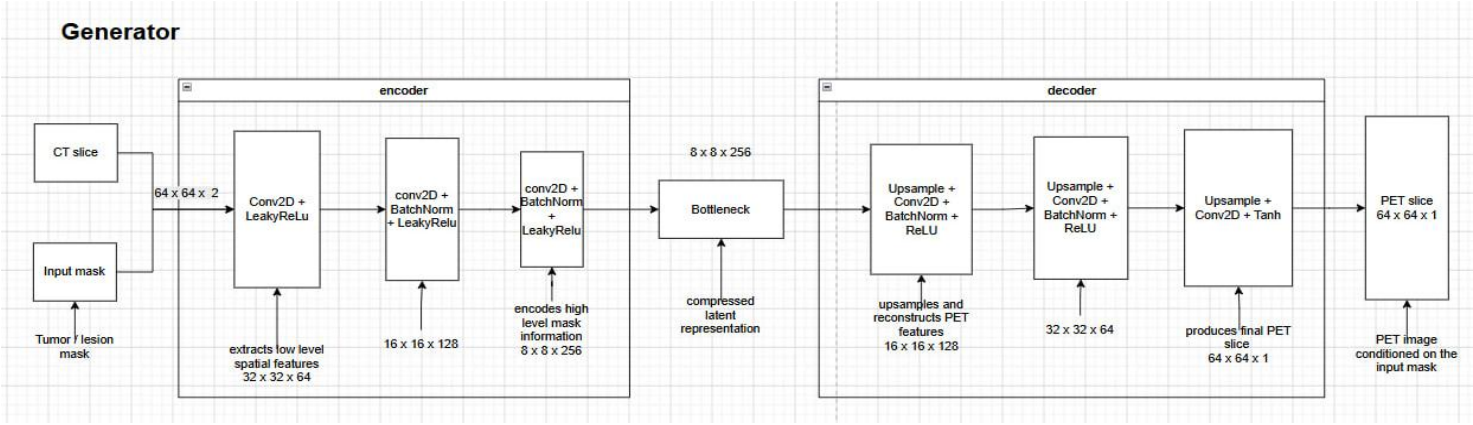


Figure 4. Generator architecture for CT + mask-conditioned PET cGAN.

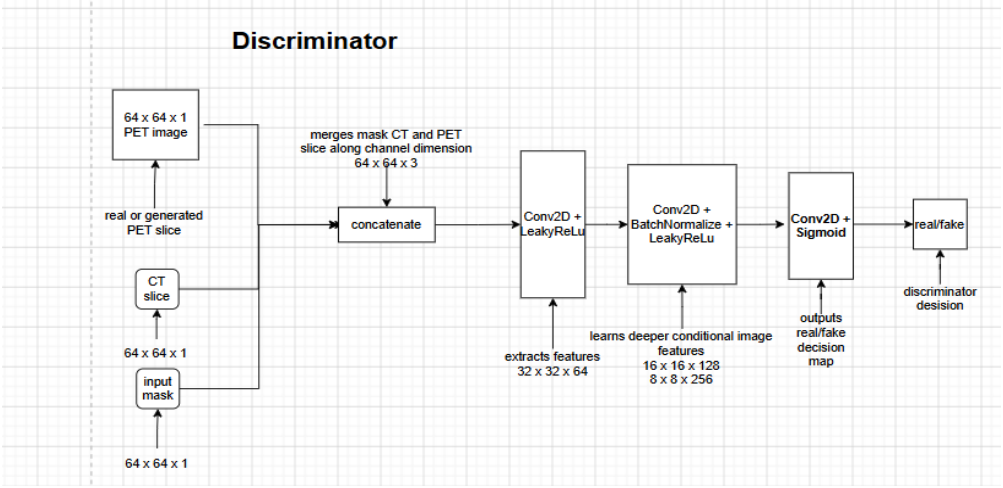


Figure 5. Discriminator architecture for CT + mask-conditioned PET cGAN.

5.8 Resize ratio and minimum mask-pixel experiments

Preprocessing experiments were conducted using the mask-conditioned PET cGAN to study how image resolution and tumor-mask visibility affect synthetic PET generation. Two parameters were varied: the image resize value and the minimum mask-pixel threshold.

The image resize value controls the spatial resolution of PET and mask slices. Smaller images reduce memory usage and training time, but may remove fine anatomical or lesion details. Larger images preserve more spatial information, but require more computation and may make training less stable. The minimum mask-pixel threshold controls how much tumor area must be visible in a slice to keep it in the dataset.

This experiment was introduced because tumor size and image resolution directly affect the information available to the model. If the tumor region is too small, the mask may not provide enough guidance. If the resolution is too low, small lesions may disappear or become too coarse. If the resolution is too high, training becomes heavier and may not necessarily improve the result.

For each preprocessing configuration, the mask-conditioned cGAN was trained using the corresponding image size and filtered slice set. The generated PET images were compared with the real PET targets using SSIM and PSNR. These metrics were used to evaluate structural similarity and reconstruction quality across preprocessing choices.

6. Results and Analysis (Appendix C)

6.1 Fashion-MNIST GAN and cGAN results

The Fashion-MNIST experiments were used as a preliminary step to understand adversarial training and the effect of conditioning. In the standard GAN, the training curves showed instability after the best epoch: the generator and discriminator losses did not converge smoothly, which is expected in adversarial learning. The generated images were partially recognizable but remained noisy and were not class-controlled. This showed the practical limitation of unconditional generation even on a simple dataset.

In the Fashion-MNIST cGAN, the losses were more stable and the generated images were more consistent with the requested class labels. Since the generator received both noise and class information, it learned not only to generate plausible samples, but also to align the output with a requested category. This confirmed the value of conditioning and motivated the transition toward conditional generation in the PET experiments, where class labels were replaced by medical conditions.

6.2 PET GAN baseline results

The PET GAN baseline was trained as an unconditional generation model. The training and validation losses evolved during training, but the generated outputs remained mostly noise-like. This shows that the loss curves alone were not sufficient to confirm useful learning: although the adversarial game continued during training, the generator did not learn a medically meaningful PET distribution.

The generated images lacked recognizable anatomical structure and did not show consistent PET-like uptake patterns. This result confirmed the main limitation of the standard GAN for this task: the generator receives only random noise and has no information about tumor location, anatomy, or patient-specific structure. Therefore, the PET GAN baseline was useful as a reference, but it was not suitable as the final approach for tumor-related PET augmentation.

6.3 PET cGAN with mask conditioning results

The mask-conditioned PET cGAN improved the generated outputs compared with the PET GAN. The loss curves still showed adversarial fluctuations, but the generated images were no longer

pure noise and began to show recognizable PET-like structure. This indicates that the tumor mask provided useful spatial guidance to the generator.

The BCE adversarial loss encouraged the generator to produce images that the discriminator would classify as realistic, while the L1 reconstruction loss encouraged voxel-wise similarity with the paired real PET slice. This combination helped preserve spatial alignment between the input mask and the generated PET image.

However, mask-only conditioning remained limited. The tumor mask gives lesion location and shape, but it does not provide anatomical context around the lesion. As a result, generated PET images could still appear blurry or incomplete.

6.4 PET cGAN with CT conditioning results

The CT-conditioned PET cGAN used anatomical information as input. Its training and validation curves showed adversarial fluctuations, but the generated images were more structured than those of the standard PET GAN. This suggests that CT provides useful anatomical guidance for PET generation.

Quantitatively, the CT-only cGAN achieved the highest PSNR and SSIM values among the compared conditioning strategies, with PSNR = 37.0096 and SSIM = 0.9632. This indicates strong pixel-level and structural similarity with the real PET images. However, CT-only conditioning does not explicitly provide lesion localization. Therefore, although the images may be close to the real PET in global structure, the generated tumor-related uptake may still be spatially inaccurate.

This result shows that PSNR and SSIM should be interpreted carefully. High numerical similarity can reflect good anatomical reconstruction, but it does not automatically guarantee that lesion-specific information is correctly generated.

6.5 PET cGAN with CT + mask conditioning results

The CT + mask cGAN combined anatomical context from CT with lesion-location information from the tumor mask. The training curves still contained fluctuations, which is normal in GAN training, but no complete mode collapse was observed because the generated outputs remained structured.

Visually, this configuration produced the most medically coherent outputs. Compared with mask-only conditioning, the generated images had better anatomical consistency. Compared with CT-only conditioning, the model also had access to explicit tumor-location information. This made the CT + mask model the most relevant configuration for tumor-focused PET generation.

Although the CT + mask cGAN did not achieve the highest PSNR, it still obtained a high SSIM value of 0.9605 and produced more meaningful outputs from a medical perspective. The slightly lower numerical score can be explained by the fact that lesion-related PET uptake is spatially sensitive: small misalignments around the tumor region can reduce PSNR and SSIM, even if the generated image is more useful for tumor-related analysis.

6.6 Resize ratio and minimum mask-pixel experiment results

The preprocessing experiments were conducted using the mask-conditioned PET cGAN to evaluate how image resolution and tumor-mask visibility affect generation quality. The experiments varied the image resize value and the minimum mask-pixel threshold.

The best quantitative result was obtained with image size 64 and minimum mask pixels 200, reaching SSIM = 0.9462 and PSNR = 37.70. This result indicates that keeping slices with sufficiently visible tumor regions improves generation quality. The line plots also showed that increasing the image size did not consistently improve SSIM, suggesting that higher resolution alone does not guarantee better generation.

However, the 64×64 resolution should be interpreted carefully. Although it performed best numerically in this experiment, small lesions may lose fine detail at low resolution. This limitation is clinically important because small lesions or early-stage micro-metastases may occupy only a few pixels after downsampling. At 64×64 resolution, such regions can become blurred, merged with surrounding background, or disappear entirely. If synthetic images are later used for data augmentation, this could reduce the usefulness of the generated data for learning small-lesion patterns. For this reason, although 64×64 produced the best quantitative score in this experiment, higher resolutions should be investigated as an immediate next step, especially for preserving small tumor structures and fine uptake boundaries. Therefore, higher-resolution experiments remain important for future work, especially if the objective is to preserve small tumor structures.

The PSNR value of 37.70 should be interpreted within the specific preprocessing experiment, where the image size and minimum mask-pixel threshold were varied using the mask-conditioned PET cGAN. This setting differed from the conditioning-strategy comparison reported in Table 1. In particular, the preprocessing experiment used a selected configuration with image size 64 and minimum mask pixels 200, while Table 1 compares the final mask-only, CT-only, and CT + mask cGAN models under their respective training settings. Therefore, the PSNR = 37.70 value and the PSNR = 36.9367 value for the mask-only model in Table 1 do not represent the same experimental condition and should not be compared directly.

6.7 Quantitative comparison of conditioning strategies

.Model	*Condition*	*PSNR*	*SSIM*	*Interpretation*
PET cGAN	Mask only	36.9367	0.9452	Provides lesion-location guidance, but limited anatomical context.
PET cGAN	CT only	37.0096	0.9632	Best numerical similarity; captures anatomy better, but lacks explicit tumor localization.
PET cGAN	CT + mask	35.3028	0.9605	High SSIM; visually more coherent because it combines anatomy and lesion guidance.

Table 1. Quantitative comparison of cGAN conditioning strategies

The comparison shows that each conditioning strategy contributes differently. Mask-only conditioning provides tumor localization but limited anatomy. CT-only conditioning provides strong anatomical structure and gives the highest PSNR/SSIM values, but it does not explicitly identify the lesion. CT + mask conditioning combines both sources of information and is therefore the most medically meaningful direction, even though it is not the best according to PSNR.

7. Project Management and Work Organization

The internship was organized as an iterative research project. Each step produced observations that motivated the next step. The work began with literature review and GAN implementation, moved toward conditional models, and then added preprocessing and evaluation experiments.

7.1 Work organization

- Bibliographic review of synthetic medical image generation, PET synthesis, and GAN/cGAN methods.
- Implementation and analysis of a standard PET GAN baseline.
- Implementation of cGAN models with mask-only, CT-only, and CT + mask conditioning.
- Preparation of figures, loss curves, and quantitative evaluation metrics.
- Continuous report and poster refinement based on supervisor feedback.

7.2 Monthly monitoring dashboard

Period	Main tasks	Deliverables
March	Literature review, understanding PET/CT data, initial GAN study	Reference list, first notes, initial notebook structure
Early April	GAN baseline and mask-conditioned cGAN implementation	GAN/cGAN notebooks, first generated outputs
Late April	CT-only and CT + mask conditioning	Updated notebooks, architecture diagrams, comparison results
Early May	Resize/minimum mask-pixel experiments, SSIM/PSNR evaluation	Metric matrices, plots, poster figures
Mid May	Report and poster finalization, defense preparation	Final report, scientific poster, GitHub references

Table 2. Project monitoring dashboard.

7.3 Professional practices

The work required a research-oriented organization: maintaining notebook versions, documenting parameter changes, saving generated outputs, and recording the interpretation of failures. The project also required regular communication with supervisors and adaptation of the report and poster based on feedback. This helped me understand the difference between writing code that runs and building a research workflow that can be explained and defended.

8. Discussion

The results show that synthetic PET generation cannot be treated as a simple image generation task. In natural image generation, visual realism is often the dominant criterion. In medical imaging, however, generated images must preserve meaningful structures and must be interpreted

cautiously. This is especially important in PET, where functional uptake patterns can be clinically meaningful.

The standard GAN baseline confirmed the limitation of uncontrolled generation. The model attempted to learn the distribution of PET images from noise but produced noise-like outputs. This does not invalidate GANs as a general method, but it shows that for this task the generation process needs additional medical guidance.

The mask-conditioned cGAN added lesion-location information and produced more meaningful PET-like outputs. However, because the mask alone does not contain anatomical context, generated images could remain blurry or incomplete. The CT-conditioned cGAN provided stronger anatomical structure and achieved the highest quantitative similarity scores, but lacked explicit tumor localization. The CT + mask cGAN combined both types of information and produced the most medically coherent visual outputs.

The metric paradox is a central finding of the project. CT-only conditioning achieved the best PSNR and SSIM because the model can reproduce dominant anatomical structures that are consistent across PET/CT pairs. The CT + mask model is more difficult to optimize because lesion boundaries and uptake regions are small and spatially sensitive. Small misalignments can reduce pixel-level metrics even when the output is more useful for tumor-focused generation.

Therefore, the evaluation of synthetic medical images should not rely only on pixel-level metrics. Visual inspection, expert review, task-based validation, and checks against memorization are all necessary for future work. Synthetic PET images should not be used directly for clinical diagnosis. They should be considered controlled research outputs until validated carefully.

9. Conclusion and Perspectives

The internship demonstrated that conditional generation is a more suitable direction than unconditional GAN generation for synthetic PET image generation. The standard GAN baseline lacked structural control, while the cGAN approaches allowed the generation process to be guided by medical information.

Among the tested directions, CT-only conditioning achieved the highest numerical scores, while CT + mask conditioning was the most medically meaningful direction because it combines anatomical context and lesion localization. This result highlights the importance of interpreting metrics carefully in medical image synthesis.

The internship allowed me to acquire practical knowledge of PET/CT preprocessing, adversarial model training, conditional image generation, evaluation metrics, and research methodology. It also strengthened my ability to analyze failures, compare methods, and present scientific results in both written and visual formats.

Future work should focus on improving training stability and output quality by testing more stable GAN variants, such as WGAN-GP or pix2pix-style cGAN models. The evaluation should also be extended to more PET/CT cases and include expert visual assessment to confirm that the generated images are medically relevant, not only visually plausible. A downstream validation protocol could also compare a segmentation model trained with real PET only against the same model trained with real plus synthetic PET, tested only on real PET data.

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11. Glossary

Term	Definition
PET	Positron Emission Tomography; a functional imaging modality that measures radiotracer uptake.
CT	Computed Tomography; an anatomical imaging modality that provides structural information.
GAN	Generative Adversarial Network; a model composed of a generator and discriminator trained adversarially.
cGAN	Conditional GAN; a GAN that receives additional information to guide generation.
BCE	Binary Cross-Entropy; adversarial classification loss used for real/fake decisions.
L1 loss	A reconstruction loss that penalizes absolute differences between generated and real images.
PSNR	Peak Signal-to-Noise Ratio; metric used to evaluate pixel-level reconstruction quality.
SSIM	Structural Similarity Index; metric used to compare structural similarity between images.
NIfTI	Neuroimaging Informatics Technology Initiative file format used for medical volumes.
Data augmentation	Techniques used to increase the amount or diversity of training data.
Mask	A binary segmentation map indicating the tumor or lesion region.
Conditioning	The use of additional input information to guide the generation process.

Appendices

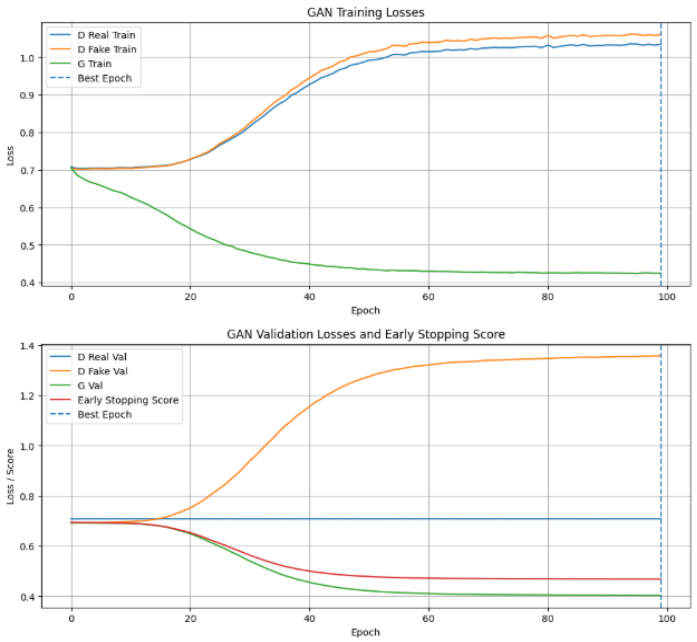
Appendix A - Technical annex: notebook deliverables

Notebook	Role
PET_GAN.ipynb	Standard GAN baseline for PET generation.
PET_mask_cGAN.ipynb	PET cGAN conditioned on tumor masks.
PET_CT_cGAN.ipynb	PET cGAN conditioned on CT images.
PET_CT_mask_cGAN.ipynb	PET cGAN conditioned on CT images and tumor masks.
PET_cGAN_experiments.ipynb	Resize and minimum mask-pixel experiments with SSIM/PSNR evaluation.

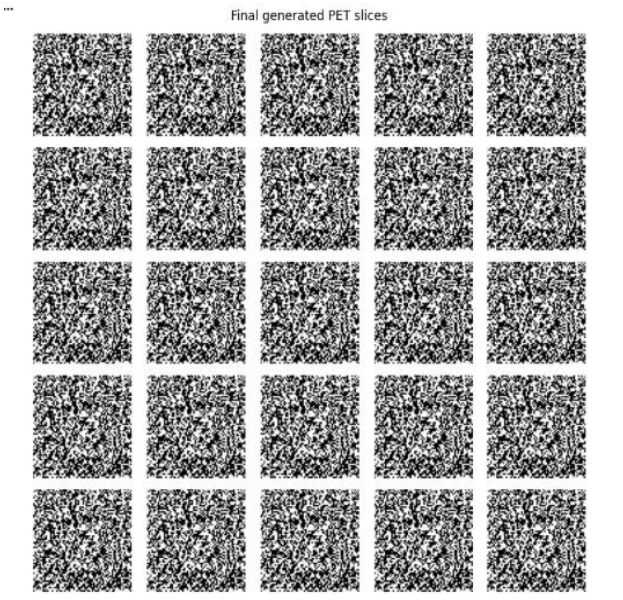
Appendix B – GitHub Repository

GitHub repository: https://github.com/LinBabayan/SyntheticPetGeneration

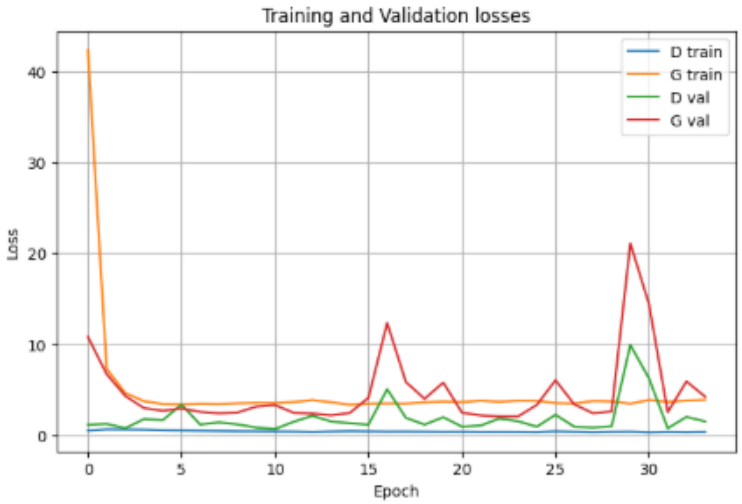
Appendix C - Additional result figures



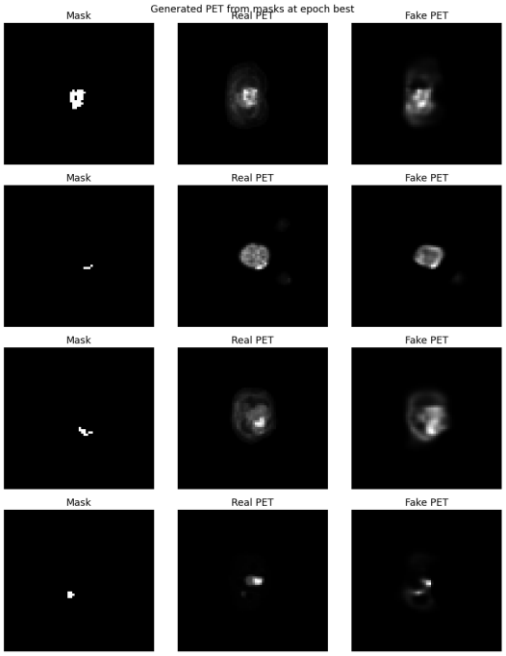
Appendix figure C1. PET GAN training and validation losses.



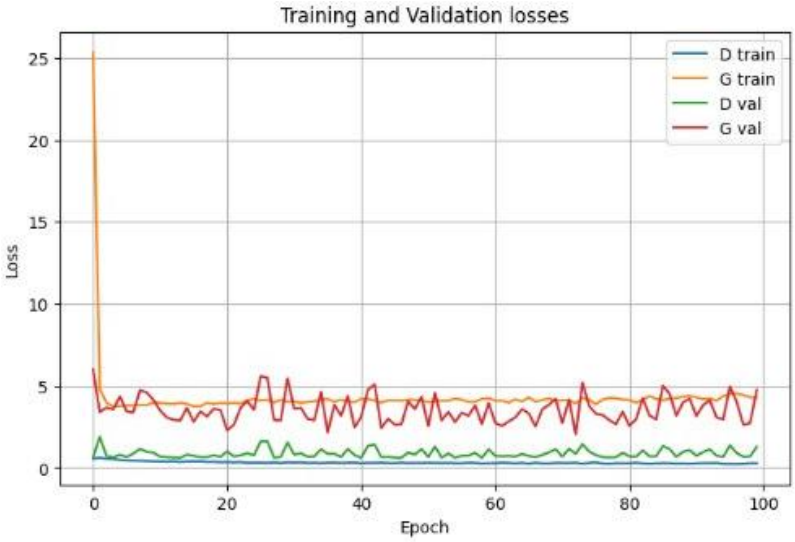
Appendix figure C2. Final PET GAN output. The generated images are mainly noise-like and lack medically meaningful structure.



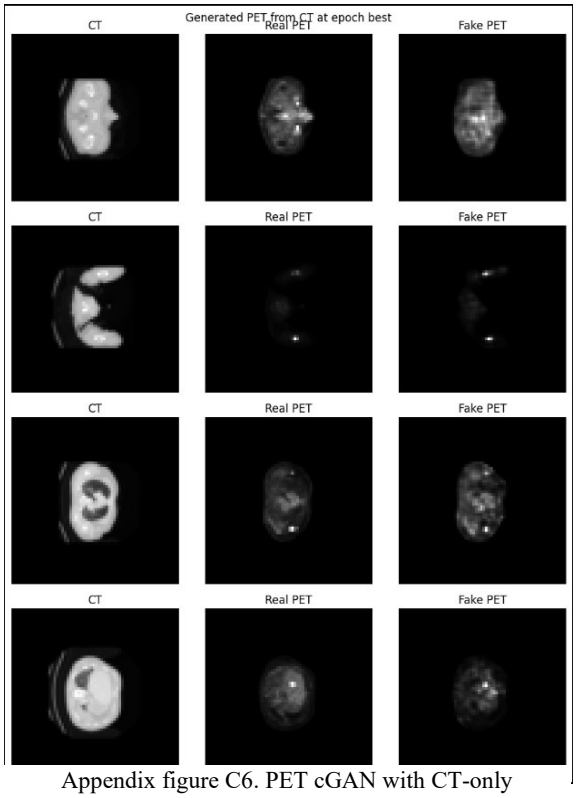
Appendix figure C3. PET cGAN with mask-only conditioning: training and validation losses.



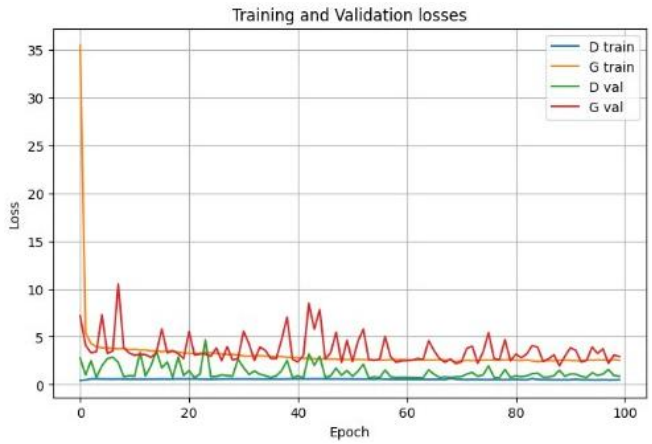
Appendix figure C4. PET cGAN with mask-only



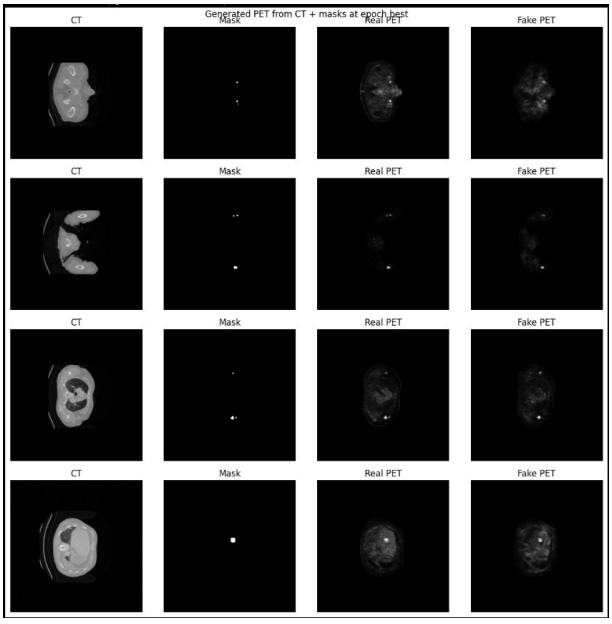
Appendix figure C5. PET cGAN with CT-only conditioning: training and validation losses.



Appendix figure C6. PET cGAN with CT-only conditioning.



Appendix figure C7. PET cGAN with CT +mask conditioning training and validation losses.



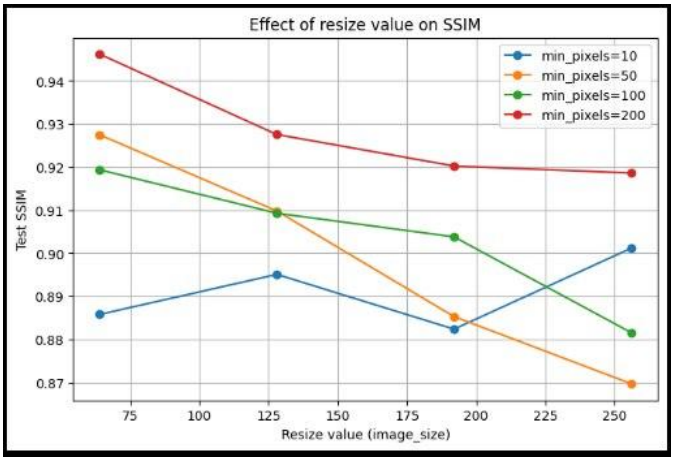
Appendix figure C8. PET cGAN with CT + mask conditioning

min_mask_pixels	10	50	100	200
image_size				
64	0.885769	0.927443	0.919328	0.946165
128	0.895060	0.909780	0.909256	0.927517
192	0.882361	0.885270	0.903768	0.920211
256	0.901146	0.869689	0.881542	0.918585

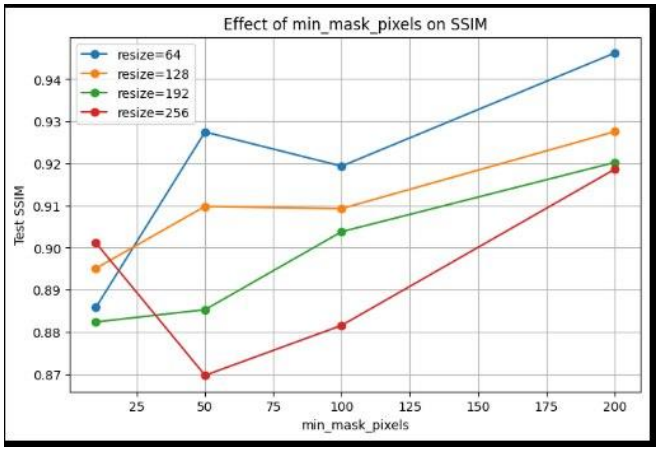
Appendix figure C9. SSIM comparison for different image sizes and minimum tumor-mask pixel thresholds.

min_mask_pixels	10	50	100	200
image_size				
64	32.321312	35.276332	34.347313	37.701952
128	31.935609	32.891377	33.184361	35.105272
192	29.628686	31.860298	31.717210	34.901859
256	30.811237	30.269782	30.634713	34.153528

Appendix figure C10. PSNR comparison for different preprocessing configurations.



Appendix figure C11. Effect of image resize value on SSIM for different minimum mask-pixel thresholds.



Appendix figure C12. Effect of minimum tumor-mask pixels on SSIM for different image sizes.

Appendix D - Repository structure and reproducibility notes

A public repository was prepared to organize code and selected references without exposing the medical dataset. The data itself was not uploaded because medical imaging data may be subject to privacy, licensing, and file-size restrictions. The repository is intended to provide the notebooks, documentation, and references necessary to understand the methodological workflow.

Repository element	Purpose
notebooks/	Contains the GAN and cGAN notebooks corresponding to the main implementation variants.
results/generated_images/	Stores selected generated outputs used in the report and poster.
results/loss_curves/	Stores training and validation loss plots.
results/metrics/	Stores PSNR/SSIM tables and preprocessing experiment matrices.
docs/	Contains the report, poster, and selected architecture diagrams if shared.
references.md	Contains the selected references used in the literature review.

Appendix Table D1. Suggested repository organization.

For reproducibility, each notebook should indicate the expected local dataset structure. A typical organization is a data folder containing image volumes and label volumes. The notebooks can then read local NIfTI files without placing the medical data in the repository.

The repository should also include a short README explaining the objective of the project, the difference between GAN and cGAN variants, the conditioning strategies, and the fact that the dataset is not included. This ensures that the work remains understandable while respecting confidentiality constraints.

Appendix E - Ethical and scientific safeguards

Synthetic medical images require careful interpretation. Even when generated images appear realistic, they should not be treated as clinical data or used for diagnosis without validation. In this internship, generated PET images were considered research outputs used to study data augmentation strategies, not clinical outputs.

A key risk of generative methods is hallucination. In medical imaging, a hallucinated uptake region could be misleading if interpreted as a real lesion. For this reason, a future clinical pipeline would need explicit safeguards, such as marking all generated data as synthetic, excluding synthetic images from direct diagnosis, and using expert review before any downstream use.

Another risk is memorization. If a model simply reproduces training slices, the generated data would not add meaningful diversity. Future validation should therefore compare generated samples with nearest training examples using similarity metrics or feature-space distances. This can help determine whether the model generates new variations or merely copies existing data.

The project also demonstrates that numerical metrics must be interpreted with caution. CT-only conditioning obtained the highest PSNR and SSIM, but CT + mask conditioning was more meaningful from the perspective of lesion localization. This shows that medical relevance cannot be reduced to a single numerical score.

Appendix F - Suggested downstream validation protocol

A downstream validation protocol could be designed to evaluate whether synthetic PET images improve a segmentation task. This was not implemented as part of the internship, but it represents an important perspective for future work.

Step	Description
Step 1	Train a segmentation model using only real PET images.
Step 2	Evaluate this model on a held-out set of real PET images.
Step 3	Train the same segmentation architecture using real PET images combined with generated PET images.
Step 4	Evaluate the augmented-data model on the same held-out real PET test set.
Step 5	Compare Dice score, IoU, precision, recall, and false positive rate.

Appendix Table F1. Proposed downstream segmentation validation protocol.

The important point is that both models must be tested only on real PET images. This avoids evaluating the segmentation model on synthetic images and makes the comparison more meaningful. If the real + synthetic training setup improves segmentation on real test data without increasing false positives, the generated data can be considered useful for augmentation.

A conservative success criterion could be a consistent improvement in segmentation metrics across several validation runs. Because medical datasets are small, a single numerical improvement is not sufficient; the evaluation should also include qualitative review of segmentation masks and error cases.

Appendix G - Personal learning outcomes

This internship contributed to both technical and professional development. Technically, I deepened my understanding of medical image preprocessing, NIfTI data handling, adversarial training, conditional generation, and quantitative evaluation using PSNR and SSIM. I also learned that model comparison in medical imaging requires careful analysis of both visual quality and medical meaning.

Professionally, the internship required autonomy and the ability to organize a research workflow. I had to read scientific papers, identify a relevant gap, implement several variants, interpret failures, and communicate the reasoning behind each methodological change. The progression from GAN to mask-conditioned cGAN, then to CT-conditioned and CT + mask-conditioned cGANs, reflects this iterative research process.

A major personal achievement was learning to treat negative results as useful information. The PET GAN baseline produced poor outputs, but this failure clarified why conditional information was necessary. Similarly, the difference between CT-only and CT + mask results showed that quantitative metrics and medical relevance do not always point to the same conclusion. This understanding is important for future work in medical artificial intelligence.

Appendix H - Implementation Problems and Solutions

The table below summarizes the main implementation difficulties encountered during the project and the corresponding solutions adopted. These issues guided the methodological progression from an unconditional GAN baseline toward conditional and multimodal PET generation.

Problem encountered	**Cause / explanation**	**Solution applied**
Many PET/CT slices contained no tumor or very small tumor regions	These slices provided weak or no lesion information for conditional generation	Applied a minimum mask-pixel threshold to keep only tumor-relevant slices
Mask boundaries became blurred after resizing	Standard interpolation created intermediate values in binary masks	Used nearest-neighbor interpolation for masks

Problem encountered	Cause / explanation	Solution applied
Standard PET GAN generated noise-like images	The generator received only random noise and no medical condition	Used cGAN models with medical conditioning
Mask-only cGAN lacked anatomical context	The mask provides lesion location but not surrounding anatomy	Added CT as an additional conditioning input
CT-only cGAN lacked explicit lesion localization	CT provides anatomy but does not directly identify tumor position	Combined CT with tumor mask conditioning
Checkerboard artifacts appeared in generated images	Decoder upsampling / transposed convolution effects produced grid-like artifacts	Replaced transposed-convolution-based upsampling with explicit upsampling followed by convolution
Loss curves showed fluctuations	GAN training is adversarial and does not converge smoothly like supervised learning	Interpreted losses together with generated images, PSNR, and SSIM
PSNR/SSIM favored CT-only conditioning	CT anatomy is easier to reproduce numerically than lesion-specific uptake	Interpreted metrics together with medical coherence and lesion localization
64×64 gave best numerical preprocessing result but may lose fine lesion detail	Lower resolution reduces fine spatial information	Suggested testing higher resolutions as future work